

Hyunkyu Kang

khk0606@naver.com | github.com/khk0606 | South Korea
khk0606.github.io

RESEARCH INTERESTS

My research interests lie in human–scene interaction, affordance-aware motion generation, and embodied AI. I study how language, spatial relationships, and interaction history can guide context-appropriate behavior, with an emphasis on reliable data pipelines and reproducible evaluation.

EDUCATION

Chung-Ang University

School of Art and Technology

Additional studies in Data Privacy

Undergraduate

RESEARCH EXPERIENCE

CREAL Lab, Chung-Ang University

Jul 2026 – Aug 2026

AI-SW Convergence Undergraduate Researcher Program | Advisor: Prof. Taeil Jin

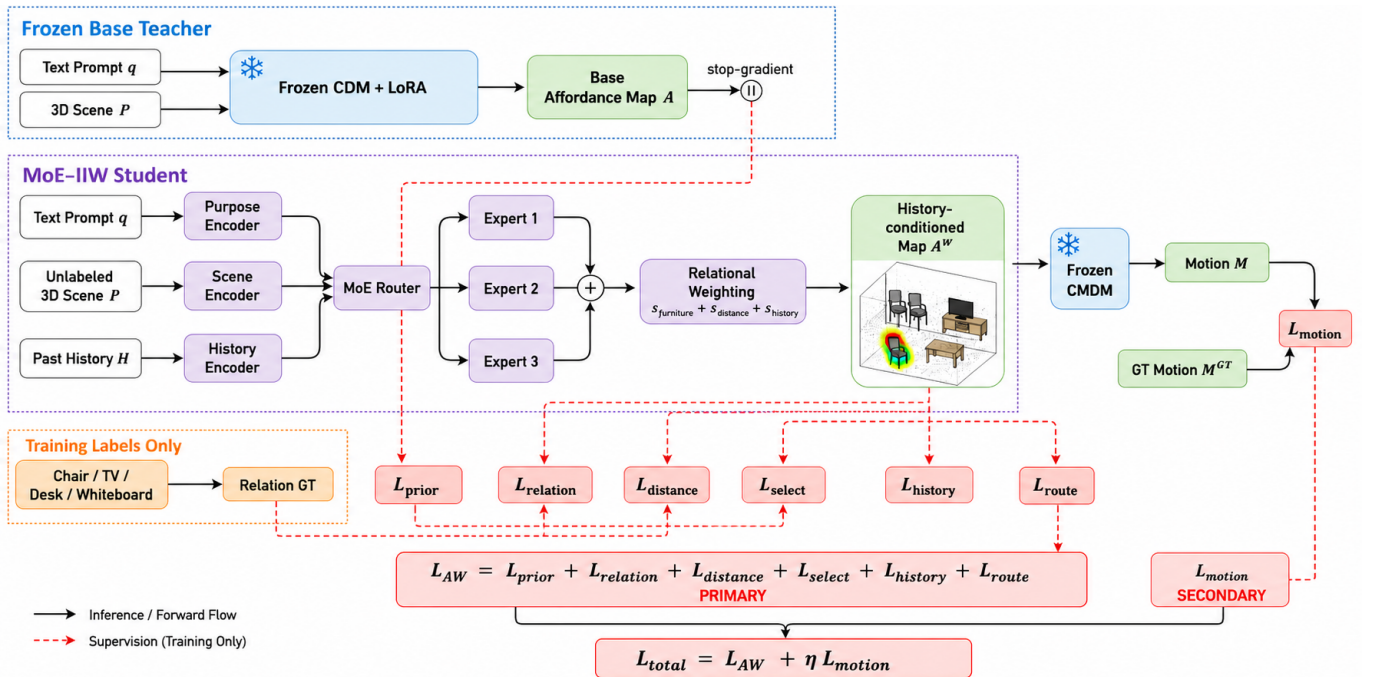
- Worked on language-based 3D scene understanding and purpose- and history-conditioned affordance maps.
- Designed a scene-point/body-part representation and implemented motion preprocessing, coordinate validation, and teacher–student training and evaluation components.

REPRESENTATIVE RESEARCH

Purpose- and History-Conditioned Affordance-Aware Motion Generation

Ongoing research | [Code](#)

- Designed an affordance representation over 8,192 scene points and six body parts, combining purpose, scene geometry, and interaction history.
- Implemented teacher adaptation, contact supervision, point-alignment checks, and checkpoint auditing within an ADM/LoRA and mixture-of-experts research workflow.
- Established strict multi-object evaluation gates and documented checkpoint limitations; end-to-end performance remains under evaluation.



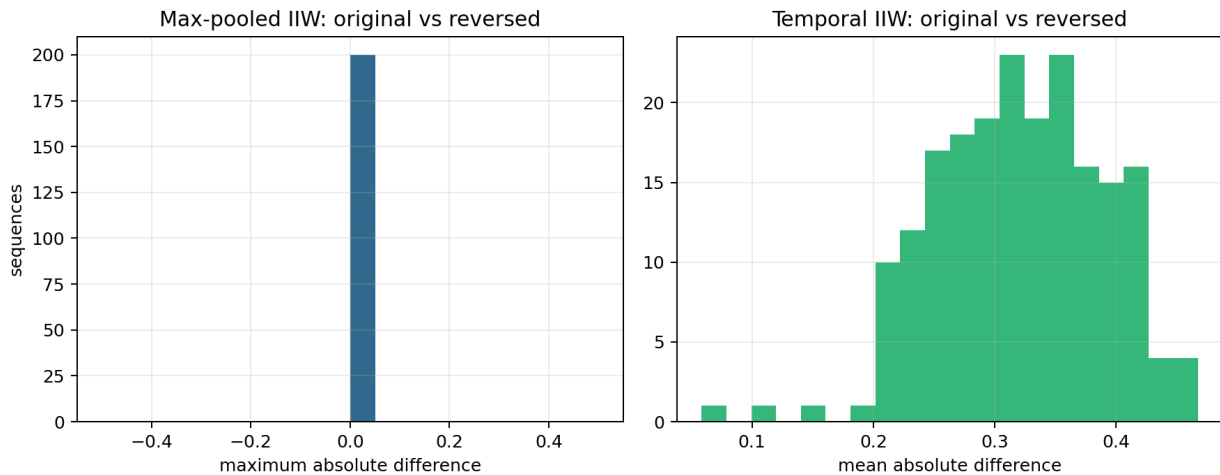
Proposed teacher–student pipeline for purpose- and history-conditioned affordances. End-to-end performance remains under evaluation.

RESEARCH AND PROJECTS

Temporal Diagnostics for Interaction Representations

Experimental analysis | Human–scene interaction

- Analyzed 200 motion sequences and their reversals: max-pooled interaction intensity weights (IIW) were identical, while temporal curves had a median mean absolute difference of 0.323.
- Characterized temporal information loss in a proxy IIW representation and compared order classifiers; temporal compression did not consistently improve classification.



Static max pooling is reversal-invariant; time-resolved curves retain order-dependent differences.

YBot-to-HumanML263 Motion Data Pipeline

Data processing and representation validation

- Built a validated 162-sequence sitting-motion dataset with natural approach trajectories and source-disjoint train/validation/test splits of 136 / 13 / 13.
- Verified motion/text/metadata consistency, zero source-split leakage, and a maximum reconstructed endpoint error of 2.27×10^{-7} m; rejected unrealistic synthetic trajectory warps.

Video-to-Robot Motion Transfer

Simulation prototype | [Code](#)

- Connected reference-video interpretation to structured motion specifications, JAX reward functions, and Dial-MPC control for a Unitree Go2 model in MuJoCo/MJX.
- Worked on startup holds, velocity ramps, warm starts, and joint-limit safeguards. Validation is simulation-based, not a physical-robot deployment.

Korean Hate-Speech Robustness under Text Obfuscation

NLP evaluation | [Code](#)

- Evaluated a balanced 500-comment K-MHaS subset across five variants (2,500 predictions per system), comparing character-based, KoELECTRA, and Qwen-based pipelines.
- Measured normalization and calibration trade-offs; normalization + character TF-IDF MLP achieved 0.772 worst-condition balanced accuracy on the evaluated benchmark.

TECHNICAL EXPERIENCE

Programming and ML: Python, PyTorch, NumPy, scikit-learn, Hugging Face Transformers

Motion and simulation: HumanML263, Unity/YBot motion processing, MuJoCo/MJX, JAX

Research workflows: Git, Linux, LoRA adaptation, experiment auditing, visualization